

TASK: Task #51635

Q.

Install the following packages and host a sample dynamic website with help of database.

- 1. Install Apache Webserver**
- 2. Install PHP**
- 3. Install MySQL/MariaDB**

Configuration files, xtensions and versions should be noticed.

1. Install Apache

```
#sudo dnf install -y httpd
```

Start Apache and enable it at boot:

```
#sudo systemctl start httpd
```

```
#sudo systemctl enable httpd
```

Check:

```
#sudo systemctl status httpd
```

Test configuration:

```
#sudo httpd -t
```

Expected:

Syntax OK

Apache configuration files:

```
/etc/httpd/conf/httpd.conf
```

```
/etc/httpd/conf.d/
```

Website directory:

```
/var/www/html/
```

2. Install PHP

```
#sudo dnf install -y php
```

Check PHP:

```
#php -v
```

Check extensions:

```
#php -m
```

Check specifically for database support:

Restart Apache:

```
#sudo systemctl restart httpd
```

Find PHP configuration:

3. Test PHP

Create/edit:

```
#sudo vi /var/www/html/index.php
```

Put:

```
<?php
phpinfo();
?>
```

Then open:

<http://YOUR-EC2-PUBLIC-IP/index.php>

For example:

<http://13.203.155.236/index.php>

Output:



PHP Version 8.3.33	
System	Linux ip-172-31-5-37.us-south-1.compute.internal 5.12.0-211.16.1.el9_2.x86_64 SMP PREEMPT_DYNAMIC Mon May 13 10:23:57 EDT 2025 x86_64
Build Date	Jul 28 2025 17:06:50
Build System	Red Hat Enterprise Linux release 10.2 (Cooper)
Build Provider	Red Hat, Inc.
Compiler	gcc (GCC) 14.3.1 20251022 (Red Hat 14.3.1-4)
Architecture	x86_64
Server API	PHP_FPM
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-apache.ini, /etc/php.d/20-fpm.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-gmp.ini, /etc/php.d/20-imagick.ini, /etc/php.d/20-intl.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mcrypt.ini, /etc/php.d/20-mysqlnd.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-pdo_mysql.ini, /etc/php.d/20-pdo_pgsql.ini, /etc/php.d/20-pdo_sqlite.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlrpc.ini, /etc/php.d/20-zip.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API20230831.NTS
PHP Extension Build	API20230831.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	provided by mbstring
Zend Max Execution Timers	disabled
IPv6 Support	enabled

4. Install MariaDB

```
#sudo dnf install -y mariadb-server mariadb
```

Start and enable:

```
#sudo systemctl start mariadb
```

```
#sudo systemctl enable mariadb
```

Check:

```
#sudo systemctl status mariadb
```

Check version:

```
#mariadb --version
```

5. Secure MariaDB

```
#sudo mariadb-secure-installation
```

6. Create the database

Enter MariaDB:

```
#sudo mariadb
```

Create database:

```
#CREATE DATABASE mywebsite;
```

Check:

```
#SHOW DATABASES;
```

Select it:

```
#USE mywebsite;
```

7. Create database user

```
#CREATE USER 'webuser'@'localhost' IDENTIFIED BY 'WebPassword123!';
```

Give permissions:

```
#GRANT ALL PRIVILEGES ON mywebsite.* TO 'webuser'@'localhost';
```

Apply:

```
#FLUSH PRIVILEGES;
```

Check permissions:

```
#SHOW GRANTS FOR 'webuser'@'localhost';
```

Exit:

EXIT;

8. Create a table

Enter MariaDB again:

```
#sudo mariadb
```

Then:

USE mywebsite;

Create table:

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(150) NOT NULL  
);
```

Insert sample data:

```
INSERT INTO users (name, email)  
VALUES  
( 'John', 'john@example.com'),  
( 'Alice', 'alice@example.com'),  
( 'Rahul', 'rahul@example.com');
```

View data:

```
SELECT * FROM users;
```

Exit:

```
EXIT;
```

9. Test database user

This is useful because it confirms that your username/password actually work:

```
#mariadb -u webuser -p
```

Enter:

WebPassword123!

Then:

```
SHOW DATABASES;
```

You should see:

mywebsite

Exit:

EXIT;

10. Install phpMyAdmin

Install php

```
#sudo yum install phpmyadmin
```

because phpMyAdmin isn't available in your enabled repositories.

Install required packages:

```
#sudo dnf install -y php php-mysqlnd php-mbstring php-json php-xml php-gd php-zip  
php-fpm wget tar
```

Go to /var/www:

```
#cd /var/www
```

Download phpMyAdmin:

```
#sudo wget https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-  
languages.tar.gz
```

Extract:

```
#sudo tar -xzf phpMyAdmin-latest-all-languages.tar.gz
```

Rename:

```
#sudo mv phpMyAdmin-*-all-languages phpmyadmin
```

Set ownership:

```
#sudo chown -R apache:apache /var/www/phpmyadmin
```

11. Configure phpMyAdmin

```
#cd /var/www/phpmyadmin
```

Copy configuration:

```
#sudo cp config.sample.inc.php config.inc.php
```

Generate secret:

```
#openssl rand -base64 32
```

Edit:

```
#sudo vi /var/www/phpmyadmin/config.inc.php
```

Find:

```
$cfg['blowfish_secret'] = '';
```

Set your generated value:

```
$cfg['blowfish_secret'] = 'YOUR-GENERATED-SECRET';
```

12. Configure Apache for phpMyAdmin

Create:

```
#sudo vi /etc/httpd/conf.d/phpmyadmin.conf
```

Add:

```
Alias /phpmyadmin /var/www/phpmyadmin
```

```
<Directory /var/www/phpmyadmin>
```

```
    DirectoryIndex index.php
```

```
    Options FollowSymLinks
```

```
    AllowOverride All
```

```
    Require all granted
```

```
</Directory>
```

Test:

```
#sudo apachectl configtest
```

Restart:

```
#sudo systemctl restart httpd
```

13. Access phpMyAdmin

in your browser:

<http://YOUR-PUBLIC-IP/phpmyadmin>

For your EC2 server, this was:

<http://13.203.155.236/phpmyadmin>

Login using:

Username: webuser

Password: WebPassword123!

Output:

The screenshot shows the phpMyAdmin web interface in a browser. The URL is 13.203.155.236/phpmyadmin/index.php?route=/database/structure&db=mywebsite. The interface is in the 'Structure' tab for the 'users' table. The table has 2 rows and 5 columns. The 'Create new table' form is visible at the bottom.

Table	Action	Rows	Type	Collation	Size	Overhead
users	Browse Structure Search Insert Empty Drop	2	InnoDB	latin1_swedish_ci	36.0 K	18

1 table Sum

Print Data dictionary

Create new table

Table name: Number of columns: Create